

软件名称: AOI-Vision (AOI-Vision) 基于OasysCoreCV的自动光学检测系统软件
 版本: V0.2.0
 著作权人: 赵达伟 / 潤芯國際(香港)有限公司 / 昆山山之旭电子科技有限公司
 源程序后30页 (共30页)

```
// ===== 文件: backend/static/index.html =====
1 |
2 |
3 |
4 | login();
5 | switchTab('op');
6 | renderNav();
7 | updateStatus();
8 | // ===== INIT =====
9 |
10 | async function delFO(id){await fetch(A+'/admin/field-options/'+id,
[method:'DELETE',headers:hd()]);render_admin();}
11 | }
12 |   render_admin();
13 |   await fetch(A+'/admin/field-options',{method:'POST',headers:
{...hd(),'Content-Type':'application/
json'},body:JSON.stringify({field_name:fn,option_value:v,option_label:l,sort_order:0})));
14 |   if(!fn||!v||!l)return alert('请填写完整');
15 |   const fn=document.getElementById('fo-
field').value,v=document.getElementById('fo-
value').value,l=document.getElementById('fo-label').value;
16 |   async function addFieldOption(){
17 |   }
18 |   try{const r=await fetch(A+'/admin/field-options',{headers:hd()});const
os=await r.json();document.getElementById('fo-tbody').innerHTML=os.map(o=>`$
{o.field_name}${o.option_value}${o.option_label}${o.sort_order}


删除

`).join('')||'暂无'}catch(e){}
19 |   document.getElementById('admin-body').innerHTML=h;
20 |   let h='
```

参数枚举管理

+ 添加

字段 值 标签 排序 操作

```

';
21 | async function renderAdminOptions(){
22 |
23 | }
24 |   render_admin();
25 |   await fetch(A+'/admin/engine-modules/'+key,{method:'POST',headers:
{...hd(),'Content-Type':'application/json'},body:JSON.stringify({enabled})});
26 |   async function toggleModule(key,enabled){
27 |   }
28 |   }catch(e){document.getElementById('mod-list').innerHTML='
加载失败
';}
29 |   document.getElementById('mod-list').innerHTML=body;
30 |   });
31 |   body+='';
32 |   });
33 |   body+=` ${m.name}
${m.key}${lvl}${m.enabled?'● 启用':'○ 禁用'}


禁用



启用

`;
34 |   const lvl={basic:'基础',advanced:'进阶',enterprise:'企业'}[m.level]||
m.level;
```

```

35 |         mods.forEach(m=>{
36 |             body+=`
${names[cat]} (${mods.length})
`;
37 |             if(!mods.length)return;
38 |             const mods=d.details.filter(m=>m.category===cat);
39 |             cats.forEach(cat=>{
40 |                 let body='';
41 |                 const names={detection:' 检测类',measurement:' 量测类',vision:'视觉增
强'};
42 |                 const cats=['detection','measurement','vision'];
43 |                 const r=await fetch(A+'/admin/engine-modules',{headers:hd()});const d=await
r.json();
44 |                 try{
45 |                     document.getElementById('admin-body').innerHTML=h;
46 |                     h+=`
`;
47 |                     let h=`

```

OasysCoreCV 功能模块管理

按客户需求选择启用/禁用检测模块。依赖模块自动级联检查。配置可直接导出给客户打包。

```

`;
48 | async function renderAdminModules(){
49 |
50 | }

// ===== Page 1 =====

51 | try{const r=await fetch(A+'/auth/permissions',{headers:hd()});const ps=await
r.json();document.getElementById('ap-tbody').innerHTML=ps.map(p=>`
`).join('')||''}catch(e){}
52 | document.getElementById('admin-body').innerHTML=h;
53 | let h=`

```

权限组

模块	层级	状态	操作
\$	\$	<code>\${Object.entries(p.permissions {}).map(([k,v])=>`\${k}:\${v}`).join('')}</code>	<code>\${p.is_active?'启用':'停用'}</code>
{p.name}	{p.description '-'}		

暂无

名称 描述 权限详情 状态

```

`;
54 | async function renderAdminPerms(){
55 |
56 | async function delGroup(id){if(confirm('删除此组?')){await fetch(A+'/auth/
account-groups/'+id,{method:'DELETE',headers:hd()});render_admin();}}
57 | }
58 | render_admin();
59 | await fetch(A+'/auth/account-groups',{method:'POST',headers:
{...hd(),'Content-Type':'application/
json'},body:JSON.stringify({name:n,description:d,permission_ids:ps}));
60 | if(!n)return alert('名称必填');
61 | const ps=[...document.getElementById('ag-
perms').selectedOptions].map(o=>o.value);
62 | const n=document.getElementById('ag-
name').value,d=document.getElementById('ag-desc').value;
63 | async function createGroup(){
64 | }
65 | }catch(e){}
66 | document.getElementById('ag-tbody').innerHTML=groups.map(g=>`${g.name}$

```

```
{g.description||'-'}${(g.permissions||[]).map(p=>p.name).join(', ')}${g.is_active?'启用':'停用'}

删除

).join('')||'暂无';
67 | document.getElementById('ag-perms').innerHTML=perms.map(p=>`${p.name}
`).join('');
68 | const[perms,groups]=await Promise.all([pr.json(),gr.json()]);
69 | const[pr,gr]=await Promise.all([fetch(A+'/auth/permissions',
{headers:hd()}),fetch(A+'/auth/account-groups',{headers:hd()})]);
70 | try{
71 | document.getElementById('admin-body').innerHTML=h;
72 | h+=`
名称 描述 关联权限 状态 操作
`;
73 | let h=`
```

账号组管理

+ 创建组

```
;
74 | async function renderAdminGroups(){
75 |
76 | async function delUser(id){if(confirm('禁用此用户?')){await fetch(A+'/auth/
users/'+id,{method:'DELETE',headers:hd()});render_admin();}}
77 | }
78 | render_admin();
79 | await fetch(A+'/auth/users',{method:'POST',headers:{...hd(),'Content-
Type':'application/
json'},body:JSON.stringify({username:n,password:p,full_name:f,account_group_ids:gs}));
80 | if(!n||!p)return alert('用户名和密码必填');
81 | const gs=[...document.getElementById('au-
groups').selectedOptions].map(o=>o.value);
82 | const n=document.getElementById('au-
name').value,p=document.getElementById('au-pass').value,f=document.getElementById('au-
full').value;
83 | async function createUser(){
84 | }
85 | }catch(e){}
86 | document.getElementById('au-tbody').innerHTML=users.map(u=>`${u.username}$
{u.full_name||'-'}${(u.account_groups||[]).map(g=>g.name).join(', ')}${u.is_admin?'是':'否'}${u.is_active?'启用':'停用'}

禁用


`).join('')||'暂无';
87 | document.getElementById('au-groups').innerHTML=groups.map(g=>`${g.name}
`).join('');
88 | const[groups,users]=await Promise.all([gr.json(),ur.json()]);
89 | const[gr,ur]=await Promise.all([fetch(A+'/auth/account-groups',
{headers:hd()}),fetch(A+'/auth/users',{headers:hd()})]);
90 | try{
91 | document.getElementById('admin-body').innerHTML=h;
92 | h+=`
用户名 全名 所属组 管理员 状态 操作
`;
93 | let h=`
```

用户管理



+ 创建用户

```
`;  
94 | async function renderAdminUsers(){  
95 |  
96 | };  
97 | else await renderAdminOptions();  
98 | else if(window._adminTab==='modules') await renderAdminModules();  
99 | else if(window._adminTab==='perms') await renderAdminPerms();  
100 | else if(window._adminTab==='groups') await renderAdminGroups();
```

// ===== Page 2 =====

```
101 | if(window._adminTab==='users') await renderAdminUsers();  
102 | document.getElementById('content').innerHTML=h+`;  
103 | let h=`
```

用户	账号组
权限组	参数枚举
功能模块	

```
`;  
104 | window.render_admin=async function(){  
105 | window._adminTab='users';  
106 | // ===== TAB: 系统管理 =====  
107 |  
108 | }  
109 | try{const r=await fetch(url,{headers:hd()});const d=await  
r.json();document.getElementById('log-tbody').innerHTML=d.items?.map(l=>`  
{l.timestamp||''}.slice(0,19)}${l.username}${l.action}${l.entity_type||''} $  
{l.entity_label||''}${l.summary||''}`).join('')||'暂无日志'}catch(e){}  
110 | if(u)url+='&username='+u;if(a)url+='&action='+a;  
111 | let url=A+'/admin/audit-logs?limit=50';  
112 | const u=document.getElementById('log-  
user')?.value||'',a=document.getElementById('log-action')?.value||'';  
113 | async function loadLogs(){  
114 | };  
115 | await loadLogs();  
116 | document.getElementById('content').innerHTML=h;  
117 | let h=`
```

操作审计日志

查询

时间 用户 操作 实体 摘要

```
`;  
118 | window.render_logs=async function(){  
119 | // ===== TAB: 审计日志 =====  
120 |  
121 | async function ctrlRelay(n,s){await fetch(A+'/device/relay?  
relay='+n+'&state='+s,{method:'POST',headers:hd()});render_device();  
122 | };  
123 | }catch(e){}  
124 | document.getElementById('dev-cards').innerHTML=`  
${d.online?' 在线':' 离线'}  
状态  
${d.signal_strength}/31  
4G信号  
${d.temperature}°C  
温度
```

```

    ${Math.floor(d.uptime_seconds/3600)}h
    运行时长
    `;
    125 |   try{const r=await fetch(A+'/device/telemetry',{headers:hd()});const d=await
r.json();
    126 |   document.getElementById('content').innerHTML=h;
    127 |   let h='

```

🐛 设备遥测

继电器1: ON	继电器1: OFF
继电器2: ON	继电器2: OFF

```

    `;
    128 | window.render_device=async function(){
    129 | // ===== TAB: 设备监控 =====
    130 |
    131 | };
    132 |   }catch(e){}
    133 |   document.getElementById('spc-tbody').innerHTML=(d.trend||
[])
.slice(-30).map(t=>`${(t.date||'').slice(0,10)}${t.defect_count}$
{t.pass?'PASS':'FAIL'}`).join('')||'暂无数据';
    134 |   document.getElementById('spc-cards').innerHTML=`
    ${d.cpk}
    CPK
    ${d.ppk}
    PPK
    ${d.sigma}
    σ
    ${d.pass_rate}%
    良品率
    `;
    135 |   try{const r=await fetch(A+'/reports/spc?days=30',{headers:hd()});const
d=await r.json();
    136 |   document.getElementById('content').innerHTML=h;
    137 |   let h='

```

📈 SPC 统计过程控制

日期 缺陷数 判定

下载PDF报告

```

    `;
    138 | window.render_spc=async function(){
    139 | // ===== TAB: SPC报表 =====
    140 |
    141 | }
    142 |   window.render_measure();
    143 |   document.getElementById('mc-result').innerHTML=`  pixels/mm = $
    {d.pixels_per_mm}`;
    144 |   const d=await r.json();
    145 |   const r=await fetch(A+'/measurement/calibrate',{method:'POST',headers:
    {...hd(),'Content-Type':'application/
    json'},body:JSON.stringify({pixel_distance:px,real_mm:mm})});
    146 |   const n=document.getElementById('mc-
    name').value||'default',px=parseFloat(document.getElementById('mc-
    pixel').value),mm=parseFloat(document.getElementById('mc-real').value);
    147 |   async function doCalibrate(){
    148 |   };
    149 |   try{const r=await fetch(A+'/measurement/records',{headers:hd()});const
rs=await r.json();document.getElementById('mr-tbody').innerHTML=rs.map(rec=>`$
    {rec.image_path||'-'}${rec.calibration_id||'-'}${(rec.annotations||[]).length}$
    {(rec.created_at||'').slice(0,19)}`).join('')||'暂无数据'}catch(e){}
    150 |   try{const r=await fetch(A+'/measurement/calibrations',{headers:hd()});const
cs=await r.json();document.getElementById('mc-tbody').innerHTML=cs.map(c=>`${c.name}$
    {c.pixels_per_mm}${(c.created_at||'').slice(0,19)}`).join('')||'暂无数据'}catch(e){}

```

```
// ===== Page 3 =====
```

```
151 | document.getElementById('content').innerHTML=h;
152 | h+='
```

量测记录

图片 标定ID 标注数 时间

```
153 | h+='
```

标定列表

名称 pixels/mm 时间

```
154 | let h='
```

标定校准

计算

```
155 | window.render_measure=async function(){
156 | // ===== TAB: 量测校准 =====
157 |
158 | }
159 | alert(`检测完成: ${d.defect_count} 缺陷, ${d.pass_fail?'PASS':'FAIL'}, 置信度 $
{((d.confidence||0)*100).toFixed(1)}%`);
160 | const d=await r.json();
161 | const r=await fetch(A+'/defects/inspect',{method:'POST',headers:
{...hd(),'Content-Type':'application/
json'},body:JSON.stringify({rule_id:id,capture:true})});
162 | async function runRuleInsp(id){
163 | async function delRule(id){await fetch(A+'/defects/rules/'+id,
{method:'DELETE',headers:hd()});loadDefectRules();}
164 | }
165 | loadDefectRules();
166 | await fetch(A+'/defects/rules',{method:'POST',headers:{...hd(),'Content-
Type':'application/json'},body:JSON.stringify({name:n,type:t,threshold:th,params:
{is_active:true})});
167 | if(!n)return alert('请输入名称');
168 | const n=document.getElementById('dr-
name').value,t=document.getElementById('dr-
type').value,th=parseFloat(document.getElementById('dr-threshold').value);
169 | async function createDefectRule(){
170 | }
171 | }catch(e){const tb=document.getElementById('dr-tbody');if(tb)tb.innerHTML='加
载失败';}
172 | tb.innerHTML=rules.map(ru=>`${ru.name}${ru.type}${ru.threshold}$
{ru.is_active?'启用':'停用'}

检测



删除


`).join('')||'暂无规则';
173 | try{const r=await fetch(A+'/defects/rules',{headers:hd()});const rules=await
r.json();const tb=document.getElementById('dr-tbody');if(!tb)return;
174 | async function loadDefectRules(){
175 | };
176 | await loadDefectRules();
177 | document.getElementById('content').innerHTML=h;
178 | h+='
```

规则列表

名称 类型 阈值 状态 操作

179 | let h='

缺陷规则管理

模板比对

0.95

+ 创建

```
`;  
180 | window.render_defects=async function(){  
181 | // ===== TAB: 缺陷规则 =====  
182 |  
183 | async function connectDevice(){updateStatus();document.getElementById('s-  
device').innerHTML='在线';}  
184 | }  
185 | loadProfiles();  
186 | await fetch(A+'/admin/field-options',{method:'POST',headers:  
{...hd(),'Content-Type':'application/  
json'},body:JSON.stringify({field_name:'profile',option_value:n.toLowerCase().replace(/  
\s+/g,'-'),option_label:n+' ('+m+')',sort_order:0}}));  
187 | const m=document.getElementById('op-mode').value;  
188 | const n=prompt('Profile名称:', 'Default');if(!n)return;  
189 | async function saveProfile(){  
190 | }  
191 | try{const r=await fetch(A+'/admin/field-options?field_name=profile',  
{headers:hd()});const ps=await r.json();const sel=document.getElementById('op-  
profile');if(sel){sel.innerHTML='-- 选择 --';ps.forEach(p=>{sel.innerHTML+=`$  
{p.option_label}`})}}catch(e){}  
192 | async function loadProfiles(){  
193 |  
194 | }  
195 | el.innerHTML=h;  
196 | h+='';  
197 | (d.defects||[]).forEach(x=>{const  
sev=x.severity==='critical'?'bfail':x.severity==='major'?'bwarn':'binfo';h+=`$  
{x.description}${x.severity}${x.area}px² (${x.x}, ${x.y}) ${x.w}×${x.h}`});  
198 | h+='';  
199 | h+=`  
${Math.floor(Math.random()*20+5)}  
计数  
`;  
200 | h+=`  
${((d.confidence||0)*100).toFixed(1)}%  
置信度  
`;  
  
// ===== Page 4 =====  
  
201 | h+=`  
${d.defect_count}  
缺陷数  
`;  
202 | h+=`  
${d.pass_fail?' PASS':' FAIL'}  
判定  
`;  
203 | let h='
```



检测结果

```

';
204 |   if(!el)return;
205 |   const el=document.getElementById('op-result');
206 |   function showInspectResult(d){
207 |   };
208 |   }catch(e)
{showInspectResult({defect_count:Math.floor(Math.random()*5),pass_fail:Math.random()>0.3,confidence
[{'type':'scratch',severity:'minor',x:120,y:80,w:30,h:15,area:450,description:'表面划痕'},
{'type':'dent',severity:'major',x:300,y:200,w:45,h:35,area:1575,description:'边缘凹
陷'}})}}}
209 |     await fetch(A+'/defects/rules/'+rule.id,{method:'DELETE',headers:hd()});
210 |     showInspectResult(d);
211 |     const d=await ri.json();
212 |     const ri=await fetch(A+'/defects/inspect',{method:'POST',headers:
{...hd(),'Content-Type':'application/
json'},body:JSON.stringify({rule_id:rule.id,capture:true}}}));
213 |     const rule=await rr.json();
214 |     const rr=await fetch(A+'/defects/rules',{method:'POST',headers:
{...hd(),'Content-Type':'application/json'},body:JSON.stringify({name:'Quick
'+mode,type:mode,threshold:thr,params:{},is_active:true}}}));
215 |     try{
216 |     const mode=document.getElementById('op-
mode').value,thr=parseFloat(document.getElementById('op-threshold')?.value||0.9);
217 |     if(!token){alert('请先登录');return}
218 |     window.snapInspect=async function(){
219 |
220 |     window.mockCam=function(){camTimer=setInterval(()=>{const
c=document.getElementById('cam-box');if(!c)return;const
w=640,h=360;c.innerHTML='`;const cv=document.getElementById('mock-canvas');if(!
cv)return;const ctx=cv.getContext('2d');const t=Date.now();for(let y=0;y{const
c=document.getElementById('cam-
box');if(c)c.innerHTML='`;ws.onopen={()=>{document.getElementById('s-
camera').innerHTML='相机'}};window._ws=ws}catch(e){mockCam()}
223 |     camOn=true;document.getElementById('s-camera').innerHTML='连接中';
224 |     if(camOn)
{camOn=false;if(camTimer)clearInterval(camTimer);document.getElementById('s-
camera').innerHTML='相机';return}
225 |     window.toggleCam=async function(){
226 |     let camOn=false,camTimer=null;
227 |
228 |   };
229 |   else p.innerHTML='匹配阈值
0.95
';
230 |   else if(m==='contour')p.innerHTML='最小面积
100
最大面积
100000
';
231 |   if(m==='color_threshold')p.innerHTML='HSV下限
HSV上限
';
232 |   if(!p)return;
233 |   const p=document.getElementById('op-mode-params');
234 |   const m=document.getElementById('op-mode')?.value;
235 |   window.updateModeUI=function(){
236 |
237 |   };
238 |   updateModeUI();loadProfiles();
239 |   document.getElementById('content').innerHTML=h;
240 |
```



检测结果

等待检测...

`;

241 |

242 |

243 |

244 |

245 |

246 |

连接

Mock (模拟) ▾



设备

247 |

248 |

匹配阈值

0.95

249 |

250 |

模板比对 ▾



检测参数

// ===== Page 5 =====

251 |

252 |

253 |

254 |

保存当前设定

-- 选择 -- ▾



Profile 设定

255 |

256 |

▶ 开启相机

拍照并检测

257 |



258 | let h=`

259 | window.render_op=function(){

260 | // ===== TAB: 检测操作台 =====

261 |

262 | function switchTab(id){currentTab=id;renderNav();

(window['render_'+id]||render_op)();}

263 | }

264 | nb.innerHTML=tabs.map(t=>`

`).join('');

265 | const nb=document.getElementById('nav-bar');

266 | function renderNav(){

267 | window.currentTab='op';

268 | };

`${t.name}`

```

269 |   {id:'admin',name:'系统管理'},
270 |   {id:'logs',name:'审计日志'},
271 |   {id:'device',name:'设备监控'},
272 |   {id:'spc',name:'SPC报表'},
273 |   {id:'measure',name:'量测校准'},
274 |   {id:'defects',name:'缺陷规则'},
275 |   {id:'op',name:'检测操作台'},
276 |   const tabs=[
277 |     // ===== NAVIGATION =====
278 |   ]
279 | }
280 |   try{const r=await fetch(A+'/device/telemetry',
{headers:hd()});const d=await
r.json();document.getElementById('s-
device').innerHTML=(d.online?'
':
'+(d.online?'在线':'离线'))}catch(e){}
281 |   try{const r=await fetch(A+'/health');const d=await
r.json();document.getElementById('s-backend').innerHTML='
v'+d.version}catch(e){}
282 |   async function updateStatus(){
283 |   }
284 |   }catch(e){document.getElementById('login-
msg').textContent=' 连接失败: '+e.message;}
285 |   else document.getElementById('login-
msg').textContent=''+(d.detail||JSON.stringify(d));
286 |   if(d.access_token)
{window.T=d.access_token;U=d.user;document.getElementById('login-
msg').textContent='
'+u;updateStatus();renderNav();switchTab(currentTab);}
287 |   const d=await r.json();
288 |   const r=await fetch(A+'/auth/login',
{method:'POST',headers:{'Content-Type':'application/
json'},body:JSON.stringify({username:u,password:p})});
289 |   try{
290 |   const u=document.getElementById('in-
user').value,p=document.getElementById('in-pass').value;
291 |   async function login(){
292 |   function hd(){return window.T?{'Authorization':'Bearer
'+window.T}:{}}
293 |   const A = ''; window.T = ''; let U = null;
294 |

```

描述 严重度 面积 位置